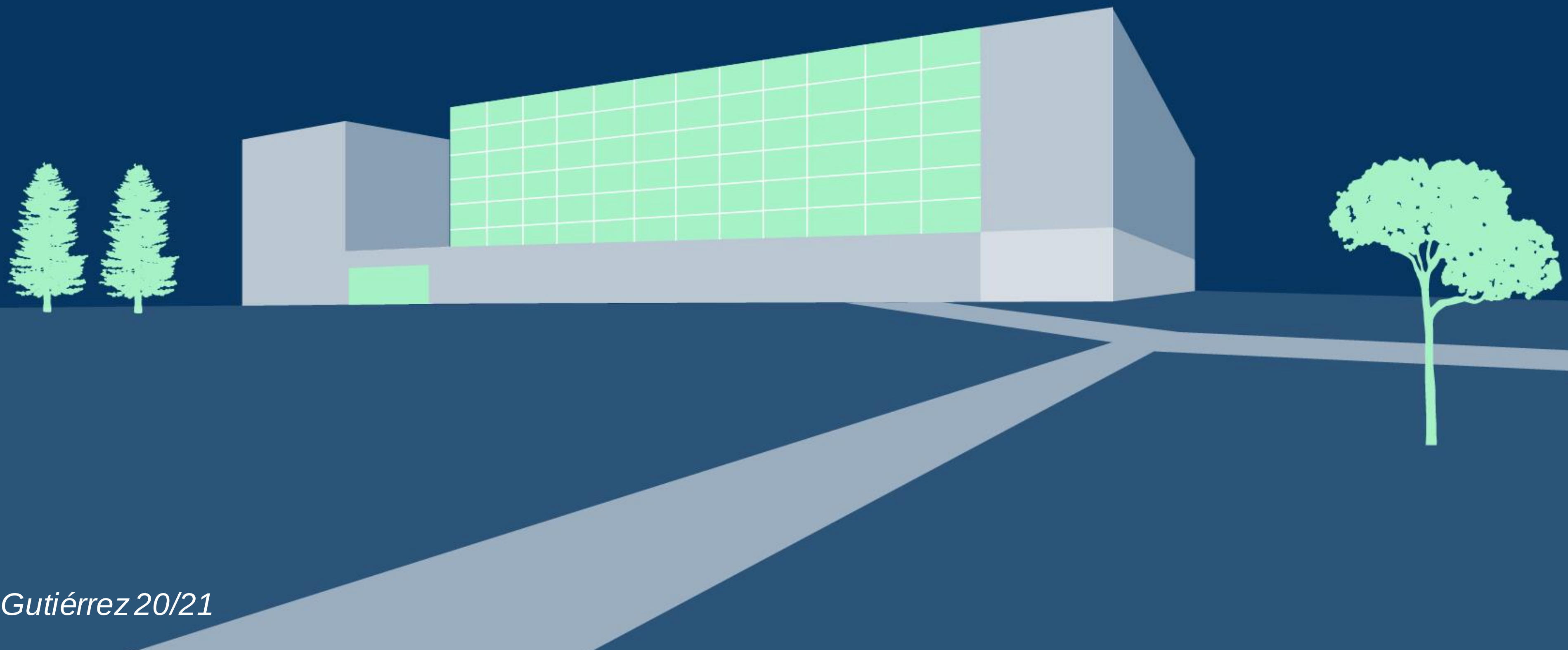
FACULTAD DE CC DE LA SALUD

Universidad Francisco de Vitoria UFV Madrid

Ciencias de la Actividad Física y del deporte

Teoría y Práctica del Entrenamiento II

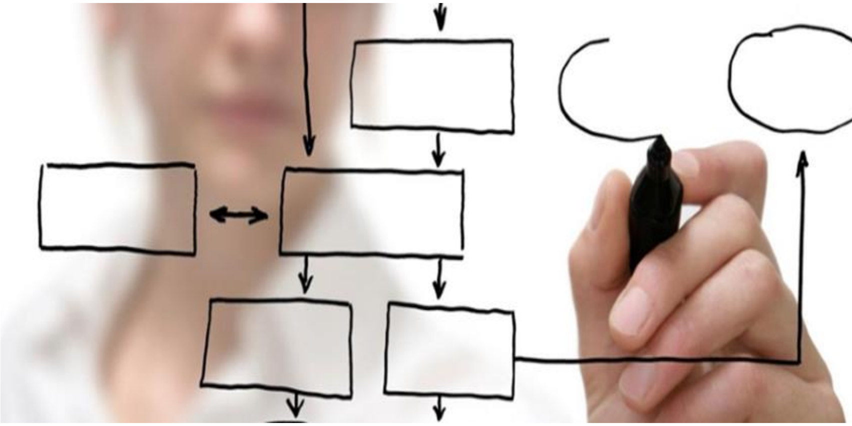


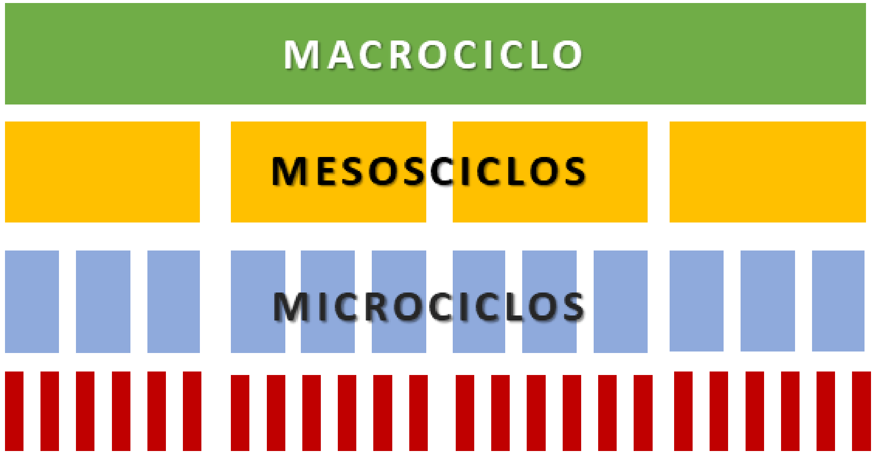
Carlos Rivera, Esther Morencos, Jorge Gutiérrez 20/21

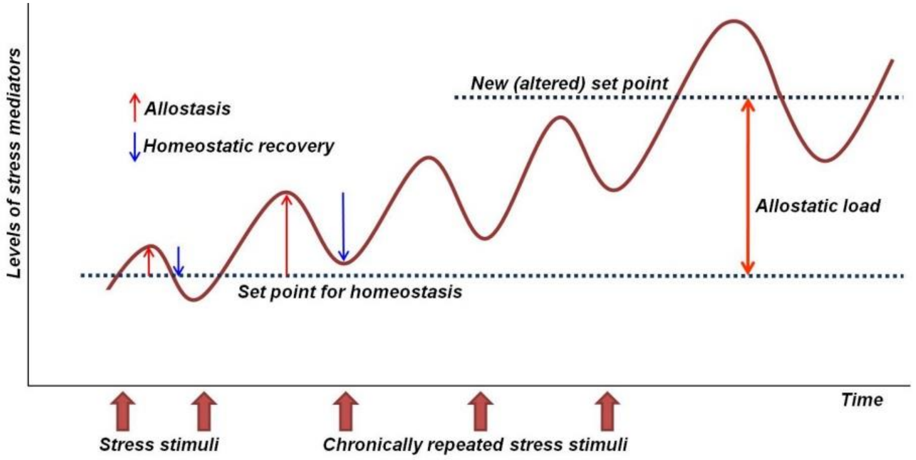
Periodización (macrociclos)

ÍNDICE

- Conceptos clave (repaso).
- Definiciones
- ¿Qué necesitamos? (primeros pasos)
- Tipos de periodización
- Visión macrociclo
- Tapering y peak performance.







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CONCEPTOS CLAVE (REPASO)

DEFINICIONES

Periodización: plan de entrenamiento que busca conseguir el máximo rendimiento variando de forma lógica y creativa los métodos de entrenamiento, intensidades y volúmenes. La periodización es un proceso teórico y práctico que permite la programación sistemática, secuencial e integradora de los entrenamientos en períodos de tiempo mutuamente dependientes para inducir adaptaciones fisiológicas específicas que aumentan el rendimiento

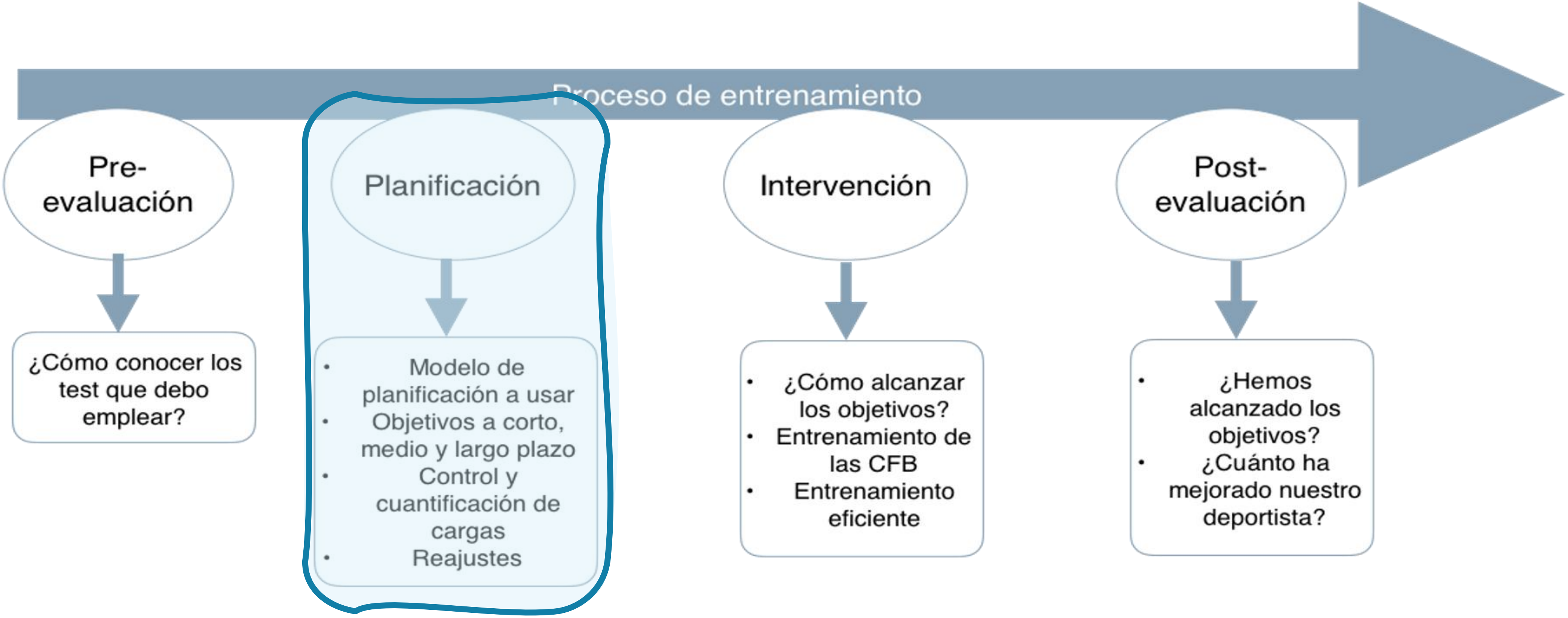
Macrociclo: es el concepto que hace referencia al plan de organización general del entrenamiento, pudiendo dividirse en: anual, bianual y olímpico.

Programación: organizar de manera concreta y al detalle todos los elementos y factores que se proponen en la planificación: objetivos, actividades, test, etc, dándoles un orden, una distribución en el tiempo y una secuenciación de acuerdo con unos criterios derivados de la teoría de entrenamiento (conocimientos procesos adaptativos).

Haff, G., & Triplett, N. T. *Essentials of strength training and conditioning*. Fourth edition 2016. Champaign, IL: Human Kinetics.

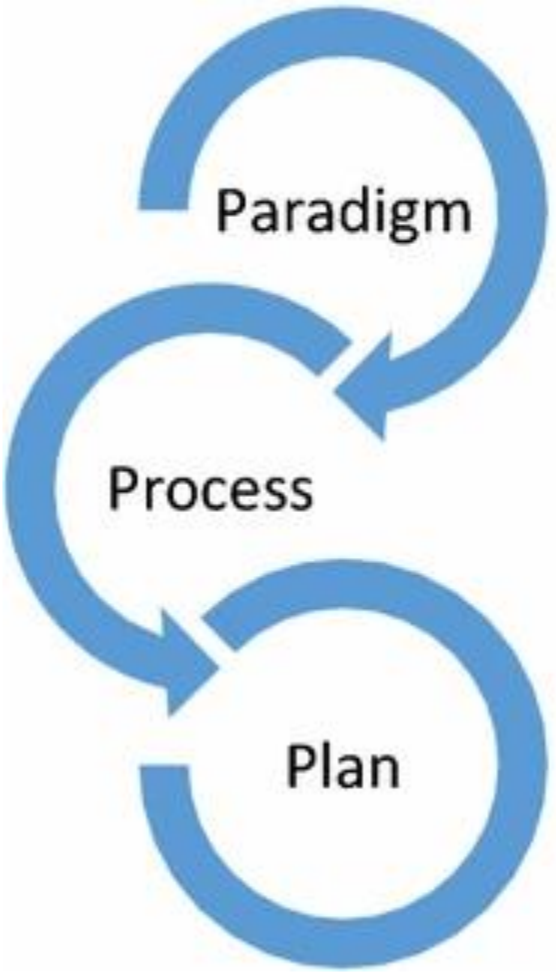
Paavo V. Komi. *Strength and Power in Sport*. 1 January 2003 International Olympic Committee

¿QUÉ NECESITAMOS? (PRIMEROS PASOS)



¿QUÉ NECESITAMOS? (PRIMEROS PASOS)

Kiely, J. Periodization Theory: Confronting an Inconvenient Truth. *Sports Med* **48,** 753–764 (2018). https://doi.org/10.1007/s40279-017-0823-y

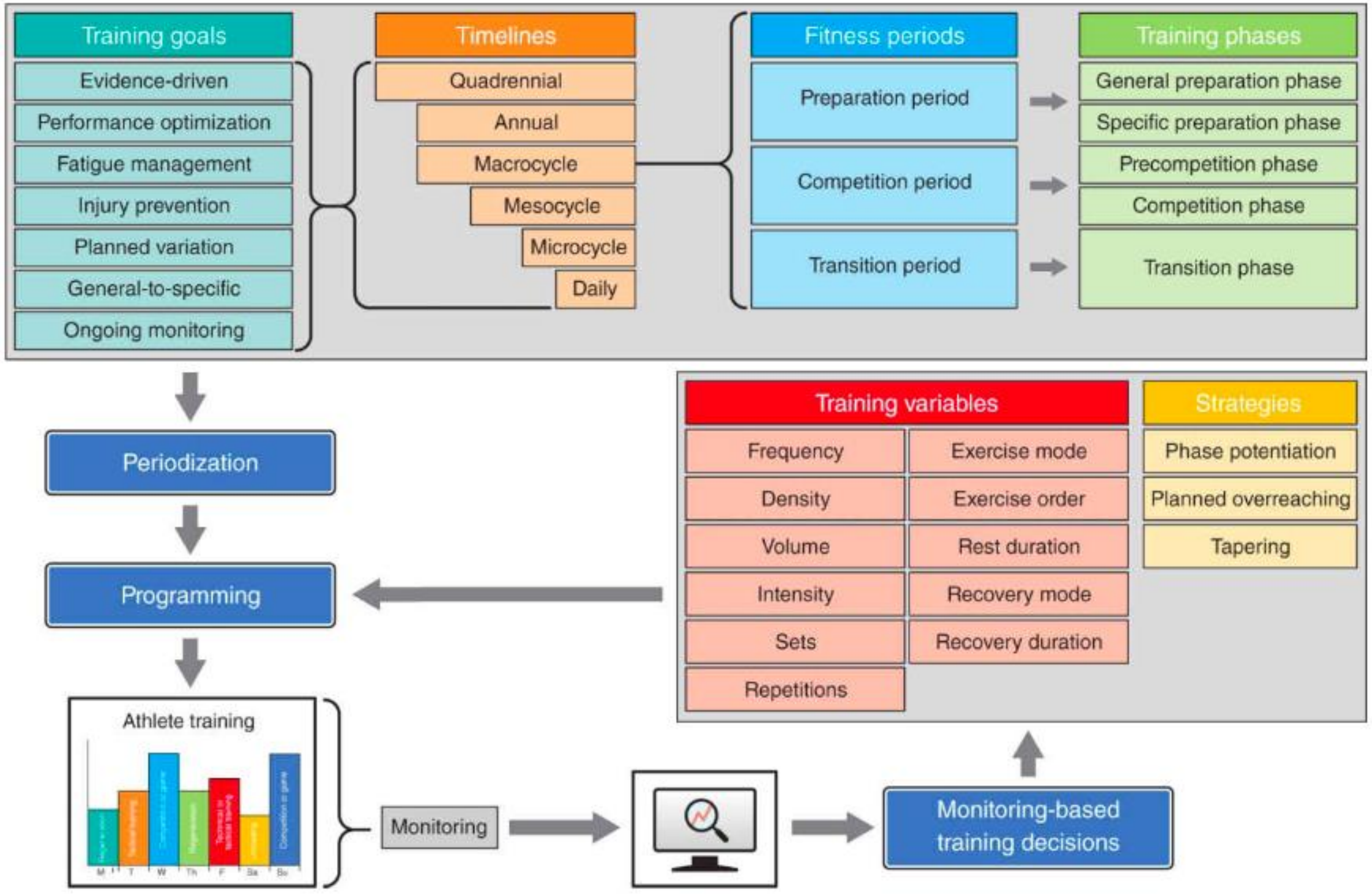


Paradigm: The coaching teams philosophical belief system, as informed by the blend of personal perspectives, critical analysis of evidence and examined experiences

Process: the set of linked procedures designed to track, analyze, review and action relevant informational outputs *(for example: subjective &/or objective monitoring data; feedback and feedforward communication flow between coaches, athletes and support team; integrated consultation, de-brief and review)*

Planning: training detail emerges under the combined influence of process outputs, integrated with the coaching paradigm, and the hard constraints imposed by logistics and competitive schedules

¿QUÉ NECESITAMOS? (PRIMEROS PASOS)



[NSCA's Essentials of Sport Science Textbook]

¿QUÉ NECESITAMOS?
(PRIMEROS PASOS)

TABLE 11.5 Basic Steps in the Planning Process

Step	Objective
1	1. Determine the athlete's long-term goals in order to develop a multiyear plan. Typically, this is accomplished with a quadrennial plan. 2. Outline the basic structure for the multiyear plan.
2	1. Prioritize the major objectives to be targeted by the annual training plan. 2. Evaluate the previous year's training plan, including competitive and performance results, and consult with the athlete or team about the training plan. 3. Create a working structure for the next annual training plan based on the competitive requirements of the athlete or team. 4. Determine the number of macrocycles contained in the annual training plan. 5. Establish the macrocycle lengths in the context of the structure established for the annual training plan.
3	1. Break the annual training plan into preparatory, competitive, and transitional phases based on the schedule of the athlete or team. 2. Divide the preparatory phases into general and specific subdivisions. 3. Create precompetitive and main competitive phases within the competitive phases of the annual training plan. 4. Insert testing days into the annual training plan at key time points.
4	1. Determine the lengths of the individual mesocycles. 2. Select and sequence the various structures of the mesocycles into the annual training plan. 3. Prioritize the focus of training factors for each mesocycle, considering how the factors are sequenced across each phase of training in the annual plan. 4. Establish the loading patterns in each mesocycle and determine how loading will progress across the macrocycles in the annual plan.
5	1. Construct each microcycle. 2. Divide the microcycle into training and recovery days according to the athlete's level of development and overall goals. 3. Establish which factors will be trained on each training day and how many training sessions will be contained during each day. 4. Create the loading structures used throughout the microcycle.
6	1. Design the individual training sessions. 2. Determine the loading structures for the training session. 3. Select the activities for the training plan.
7	1. Implement the training plan. 2. Continually monitor and evaluate the training plan and process.

Objetivos a largo plazo.
Línea básica de tiempo.

Priorizar objetivos.
Evaluar temporada anterior.
Estructura de trabajo según competiciones.
Número de macrociclos y tamaño.

Dividir en mesociclos y las correspondientes subdivisiones.
Test.

Tamaño de los mesociclos y su secuencia.
Priorizar factores entrenamiento en cada meso ciclo y su carga.

Hacer microciclos y sesiones entrenamiento.
Carga, factores, número sesiones...
Estructura del microciclo.

Diseño sesiones, carga y tareas entrenamiento.

Plan de entrenamiento.
Evaluación del plan y su proceso.

Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

¿QUÉ NECESITAMOS? (PRIMEROS PASOS)

Plan/programa

Conocer los requisitos de movimiento del deporte, las habilidades motrices y relacionarlos con las técnicas y tácticas.

Conocer el desarrollo biológico, psicológico y social del deportista.

Establecer un clima de aprendizaje adecuado con programas y estructuras adecuados.

Brewer, C. *Athletic Movement Skills Training for Sports Performance*. 2017. Human Kinetics, Inc

TIPOS DE PERIODIZACIÓN

La periodización inversa y tradicional son una estrategia eficaz para mejorar las variables biomecánicas, de rendimiento y fisiológicas durante una de carrera de 2 km, para mejorar la capacidad técnica de natación, así como el rendimiento de natación aeróbica y anaeróbica.

Además, ambos tipos de periodización de entrenamiento propuestos en este estudio mantienen los valores de composición corporal de los triatletas aficionados.

En comparación con la periodización tradicional, la periodización inversa mejora eficazmente el rendimiento del salto horizontal.

No hay recetas universales

Clemente – Suárez, J.; Ramos – Campo, D. *Effectiveness of Reverse vs. Traditional Linear Training Periodization in Triathlon.* International Journal of Environmental Research al Public Health (2019).

TIPOS DE PERIODIZACIÓN

Cada nadador parece demostrar una forma individual de responder al entrenamiento y presenta variaciones individuales

Los "picos de rendimiento" están en la mitad y al final del macrociclo debido a los mayores estímulos anaeróbicos ofrecidos y el tapering.

Los resultados deben evaluarse con cautela y deben tenerse en cuenta las características individuales del nadador y la fase de entrenamiento en la que se realizó la prueba durante la temporada. Además, los estímulos de entrenamiento ofrecidos en una fase inducirán adaptaciones fisiológicas y mecánicas de acuerdo a este tipo de estímulo, de esta manera sabremos si las adaptaciones al entrenamiento conducirán a un rendimiento adecuado en el futuro o no.

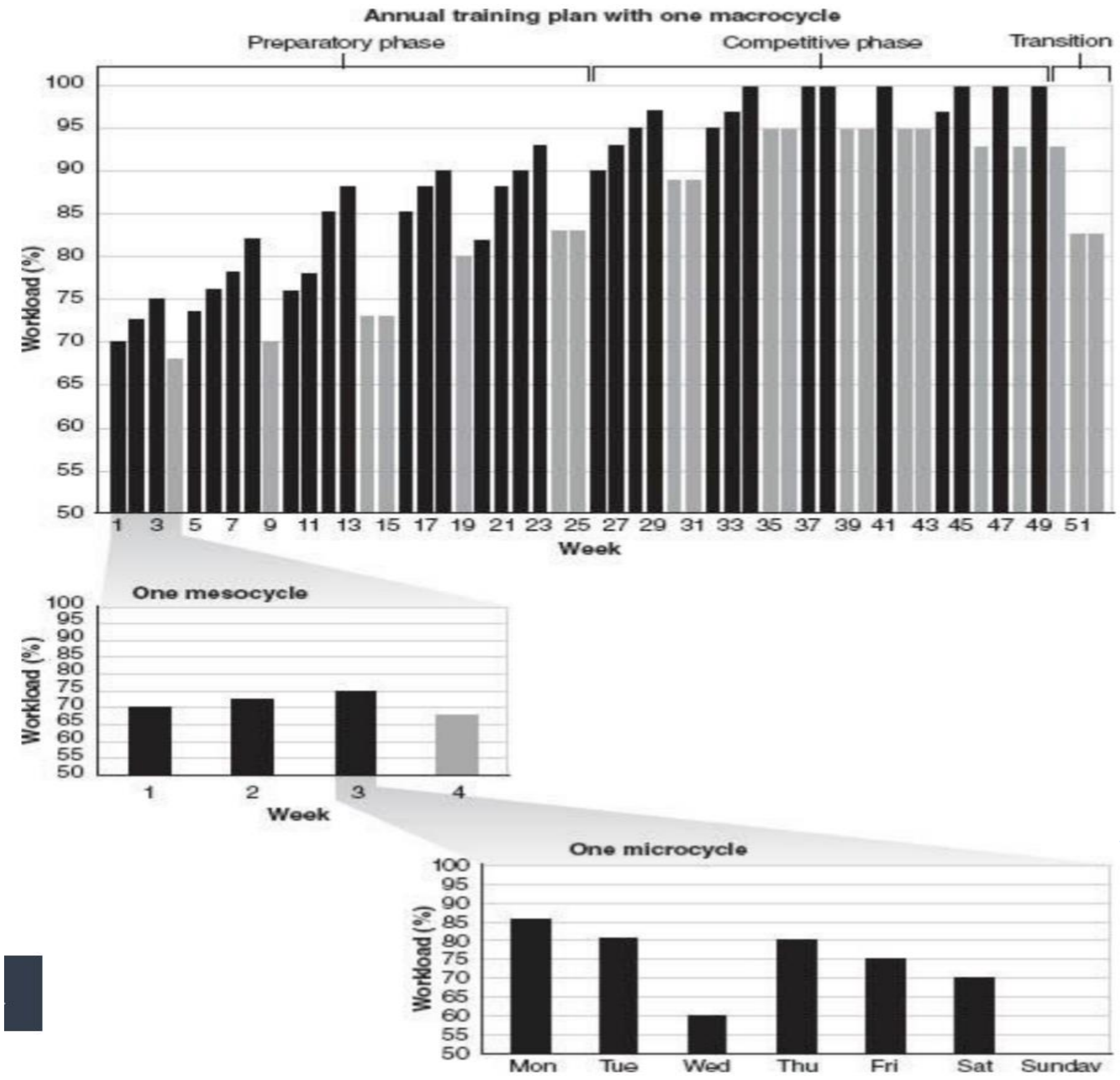
Personalización.

Adaptación.

Lógica.

Tucher, G.; De Souza, F.; Domingos Garrido, N.; Fernandes, R. *Monitoring Changes Over a Training Macrocycle in Regional Age-Group Swimmers.* Journal of Human Kinetics (2019).

VISIÓN MACROCICLO



Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

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VISIÓN MACROCICLO

Dominant (primary) training emphasis	Compatible training factors
Aerobic endurance	Strength-endurance training
	Maximal strength training
	Anaerobic endurance training
	Technical and tactical training (if done first)
Anaerobic endurance	Strength-endurance training
	Aerobic–anaerobic mixed endurance training
	Power endurance training
	Sprint (agility) training
	Explosive strength training / muscular power
	Muscular strength training
	Technical and tactical training (if done first)
Sprint ability	Maximal strength training
	Plyometric training
	Explosive strength training / muscular power
	Agility training
	Technical and tactical training (if done first)
Maximal strength	Sprint training
	Agility training
	Explosive strength training / muscular power
	Anaerobic endurance training
	Technical and tactical training (if done first)
Explosive strength/ muscular power	Sprint training
	Agility training
	Maximal strength training
	Plyometric training
	Technical and tactical training (if done first)
Technical training	Any emphasis as long as it is performed before other training factors
Tactical training	Any emphasis as long as it is performed before the other training factors

Adapted from Issurin (43, 44).

Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

MACRO – 1 CICLO.

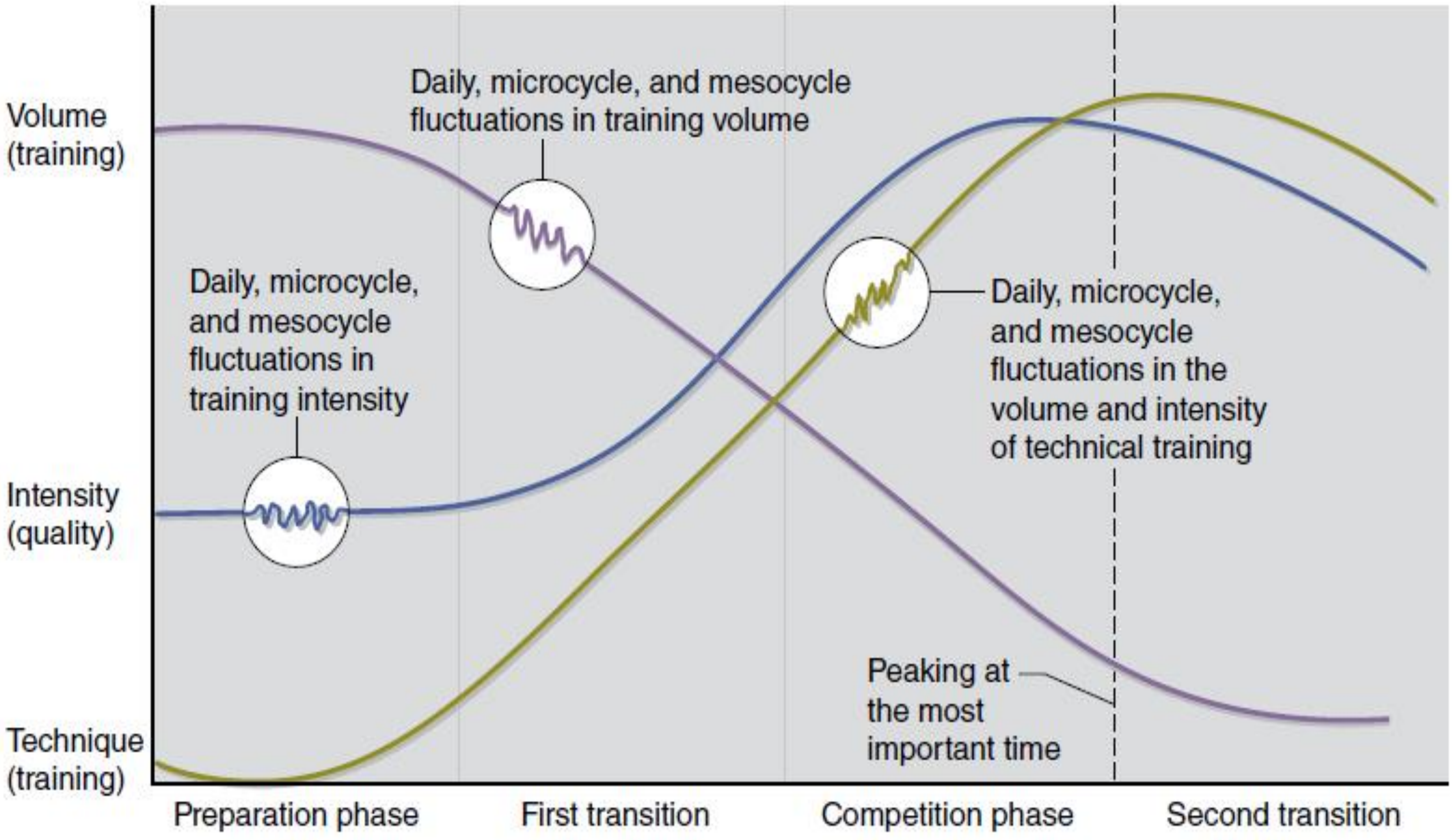

FIGURE 21.4 Matveyev's model of periodization (appropriate for novices).

Reprinted, by permission, from G.G. Haff and E.E. Haff, 2012, Training integration and periodization. In *NSCA's guide to program design*, edited by J. Hoffman (Champaign, IL: Human Kinetics), 223; adapted from figure 11.7, p. 2239. Reprinted from *Weight training: A scientific approach*, 2nd edition, by Michael H. Stone and Harold St. O'Bryant, copyright © 1987 by Burgess.

Haff, G., & Triplett, N. T. *Essentials of strength training and conditioning*. Fourth edition (2016). Champaign, IL: Human Kinetics.

MACRO – VARIOS CICLOS.

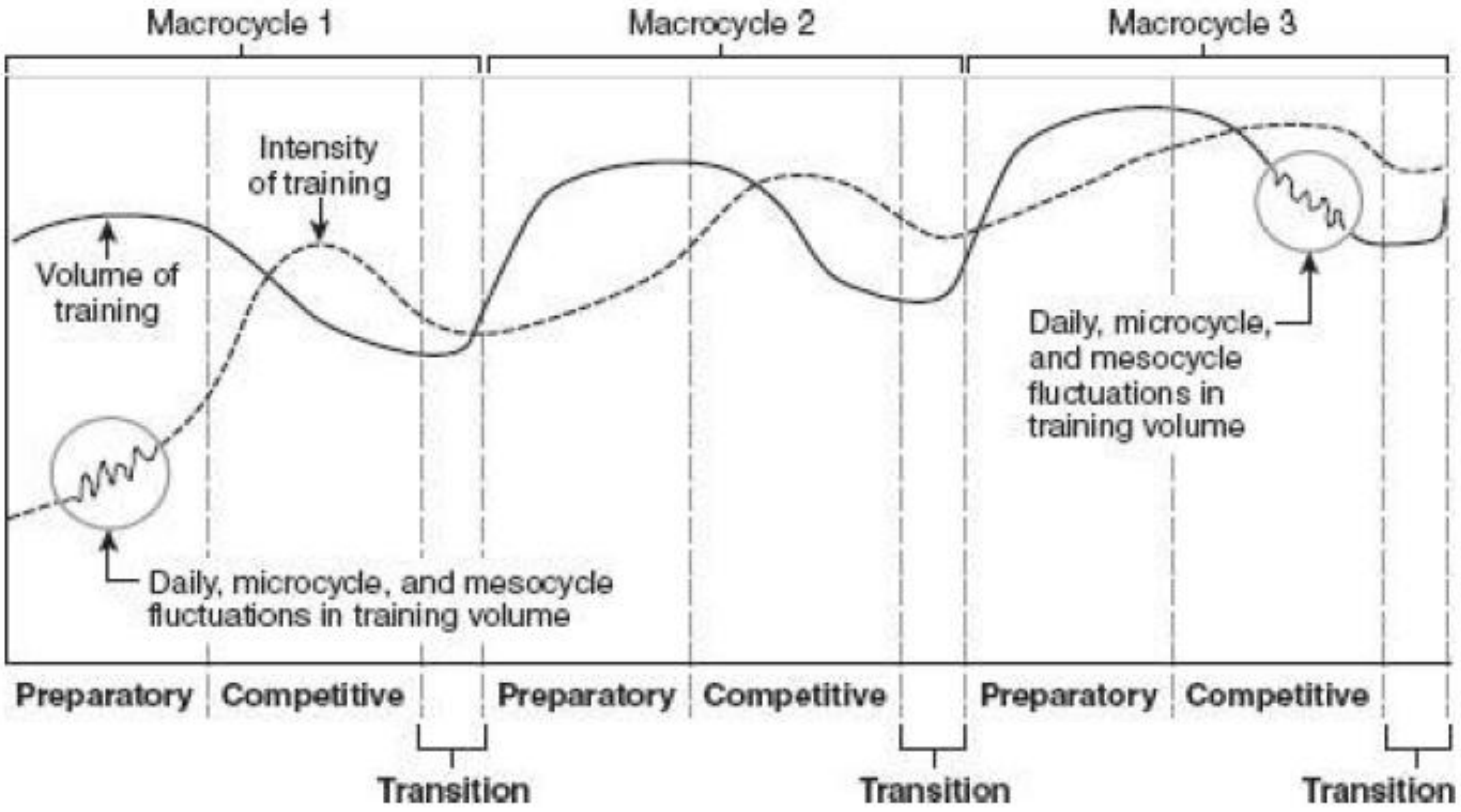

FIGURE 11.8 Example of an annual training plan with three macrocycles.
Adapted, by permission, from V. Issurin, 2008, *Block periodization:* Breakthrough in sports training, edited by M. Yessis (Ultimate Athlete Concepts), 213.

Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

MACRO.

		Annual training plan											
Month		10	11	12	1	2	3	4	5	6	7	8	9
Model	Variant												
Classic	1	Preparatory phase				Competitive phase	Preparatory phase				Competitive phase		
		General prep.		Specific prep.			General prep.		Specific prep.				
	2	Preparatory phase				Preparatory phase				Competitive phase			Transition
		General prep.		Specific prep.		General prep.		Specific prep.					
	5	Preparatory phase				Competitive phase	Preparatory phase				Competitive phase		Transition
		General prep.		Specific prep.			General prep.		Specific prep.				
	8	Transition	Preparatory phase						Competitive phase				
			General prep.			Specific prep.							
	12	Transition	Preparatory phase								Competitive phase		
			General prep.				Specific prep.						
Alternative	3	Transition	Preparatory phase		Transition	Preparatory phase		Competitive phase					
	4	Transition	Preparatory phase		Preparatory phase				Competitive phase				
	6	Transition	Preparatory phase		Competitive phase	Transition	Preparatory phase		Competitive phase				
	7	Transition	Preparatory phase		Competitive phase		Preparatory phase				Competitive phase		
	9	Transition	Preparatory phase			Transition	Preparatory phase			Competitive phase			
	10	Transition	Preparatory phase			Preparatory phase					Competitive phase		
	11	Transition	Preparatory phase			Transition	Preparatory phase				Competitive phase		
	13	Transition	Preparatory phase				Preparatory phase						Competitive phase
	14	Preparatory phase				Transition	Preparatory phase				Competitive phase		Transition
	15	Transition	Preparatory phase					Competitive phase					
	16	Transition	Preparatory phase						Competitive phase				
	17	Transition	Preparatory phase							Competitive phase			
	18	Transition	Preparatory phase								Competitive phase		

FIGURE 11.9 Example of sequences for an annual training plan based on classic and alternative models.
Adapted from A. Bondarchuk, 1986, Periodization of sports training, *Legkaya Atletika* 12:8-9.

Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

MACRO – ATR.

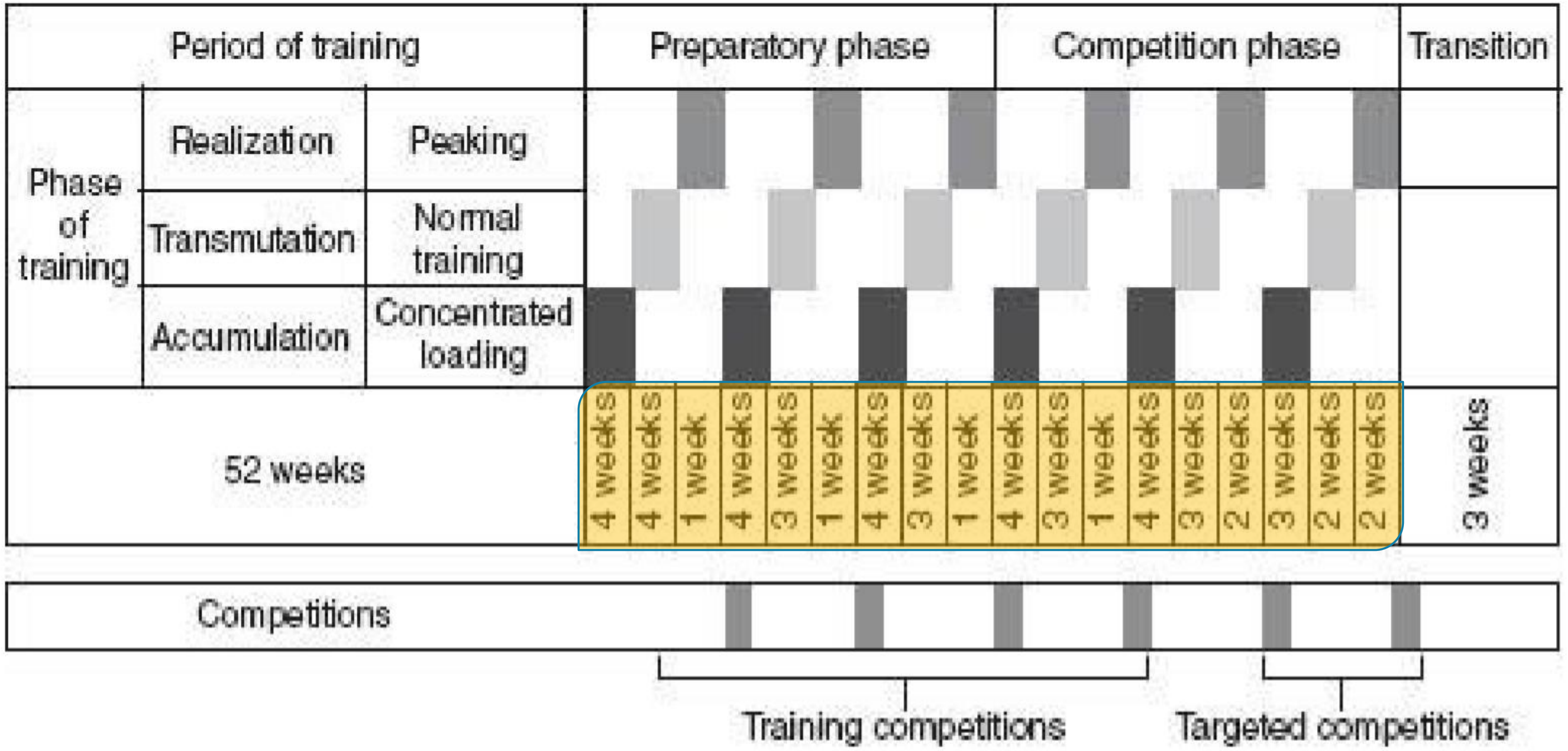

FIGURE 11.12 Example of an annual training plan containing a sequential application of accumulation, transmutation, and realization mesocycles.
Adapted by Issurin (43, 44).

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Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

MACRO – ATR.

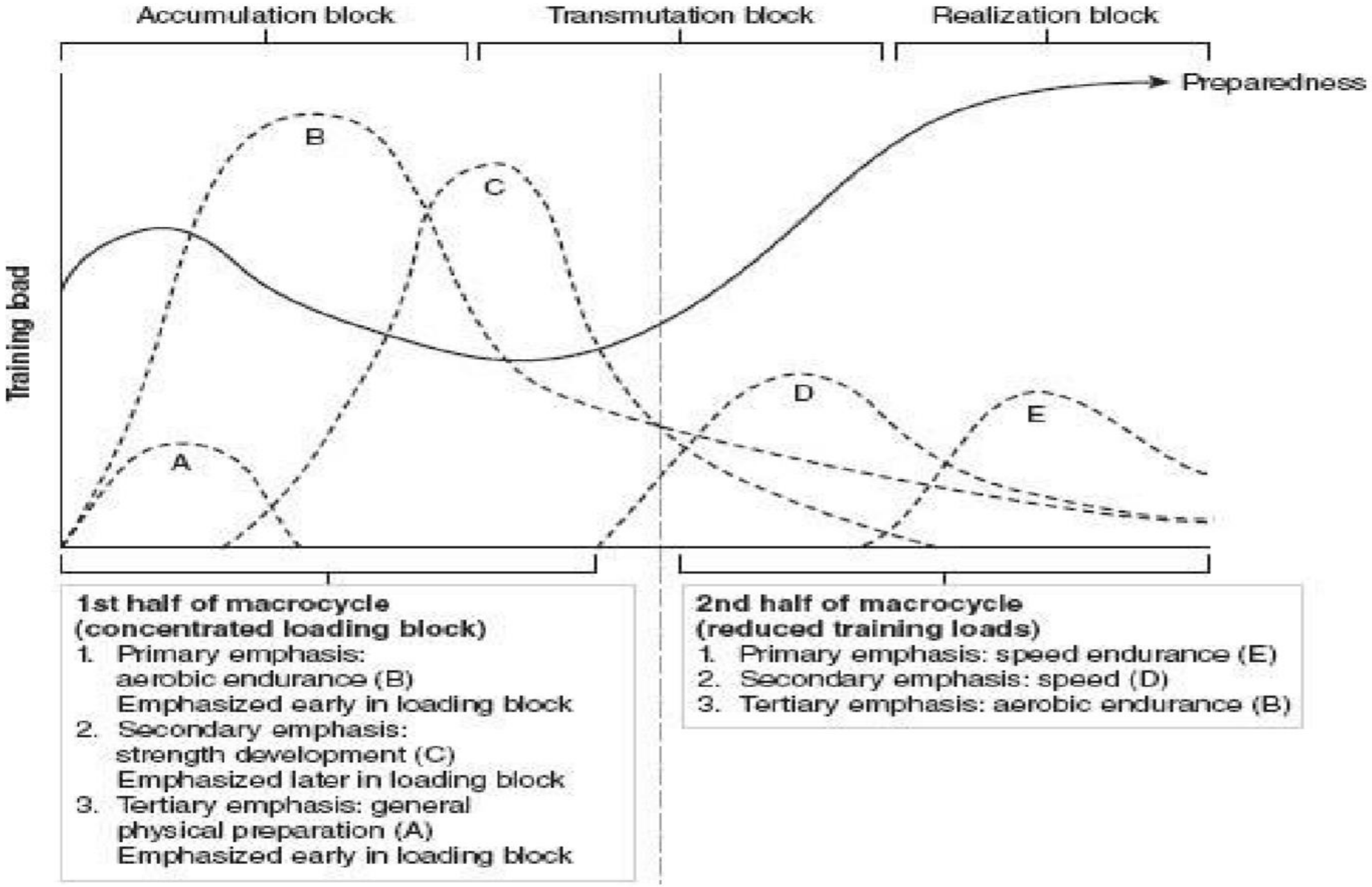

FIGURE 11.16 Sample sequencing model for a sport that requires medium-duration endurance. Adapted from Issurin (43, 44), Plisk (62), and Bompa and Haff (6).

Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

TEST.

¿Para qué?

- Evaluar al deportista(s).
- Tener valores de referencia para el entrenamiento.
- Evaluar el progreso y el proceso de entrenamiento.
- Identificar capacidades físicas a mejorar.

¿Cuándo?

- Al empezar y terminar la temporada.
- Al terminar un mesociclo.
- Cercano a la competición.
- Depende…

¿Cómo?

- Validity & reliability (que mida lo que queremos y que se pueda replicar).
- Test relacionados con los factores de rendimiento.
- Estandarización de las pruebas, protocolo y orden.

Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).
Haff, G., & Triplett, N. T. *Essentials of strength training and conditioning*. Fourth edition (2016). Champaign, IL: Human Kinetics.
Turner, A.; Comfort, P. *Advanced Strength and Conditioning*. Routledge (2018).

TAPERING.

Tapering para competir
¿Qué método de trabajo?
Diseñado por @YLMSportScience
Traducción por @AaronSanño
¿Qué es el tapering?
El tapering es "la reducción progresiva y no lineal de la carga de trabajo durante un tiempo variable cuyo objetivo es reducir el estrés físico y psicológico del entrenamiento diario y optimizar el rendimiento deportivo"
ESTRATEGIAS DE TAPERING
INTENSIDAD DEL ENTRENAMIENTO
Debe ser mantenida durante el tapering
VOLUMEN DE ENTRENAMIENTO
Máximos beneficios obtenidos con una reducción total del 41%-60% de los valores previos al tapering
FRECUENCIA DE ENTRENAMIENTO
Disminución del número de sesiones semanales no ha demostrado mejorar el rendimiento
DURACIÓN DEL TAPERING
De 8 a 14 días parece ser la línea entre la desaparición de la fatiga y el efecto negativo de desentrenarse
Las mayores ganancias en rendimiento se encuentran al efectuar una sobrecarga de entrenamiento antes del tapering. Durante este periodo, la atención deberá estar en no llegar al overreaching (sobrepasarse) pues podría afectar a la mejora del tapering
RESPUESTA INDIVIDUAL
Se han observado grandes diferencias entre atletas en la respuesta al tapering. Este método de trabajo puede ser útil para el diseño de periodización de los entrenadores, pero necesita ser personalizado individualmente para facilitar el pico de forma
Referencias: Le Meur, Hausswirth & Mujika, Tapering for competition: A review, Science & Sports, 2012

QUÉ ES EL TAPERING
ESTRATEGIAS PARA AUMENTAR EL RENDIMIENTO EN COMPETICIÓN
CROWN SPORT NUTRITION
Referencia: Le Meur, Y., Hausswirth, C., & Mujika, I. Science & Sports, 2012
El tapering es el periodo previo a una competición en el que se reduce la carga de trabajo de manera progresiva y no lineal con el fin de disminuir el estrés físico y psicológico del entrenamiento para mejorar el rendimiento deportivo
DURANTE EL TAPERING
8-14
DURACIÓN 8-14 días
INTENSIDAD mantener
VOLUMEN ↓41-60%
Nº SESIONES mantener
Las mayores mejoras en el rendimiento se producen cuando hay un aumento de la carga de entrenamiento antes del tapering.
Se debe controlar al máximo la carga interna del deportista para no provocar sobreentrenamiento.
rendimiento
intensidad
volumen
CROWN SPORT NUTRITION
diseño & edición
ACG
adrián castillo garcía

PEAK PERFORMANCE

- Peak performance
  - Objetivo final del entrenamiento.
    - Secuenciar y estructurar factores del entrenamiento.
  - Relación volumen e intensidad.
    - Tapering (7 – 28 días).
  - 7 – 14 días de promedio que se puede mantener.
    - Relación con la intensidad promedio.

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Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

MACRO.

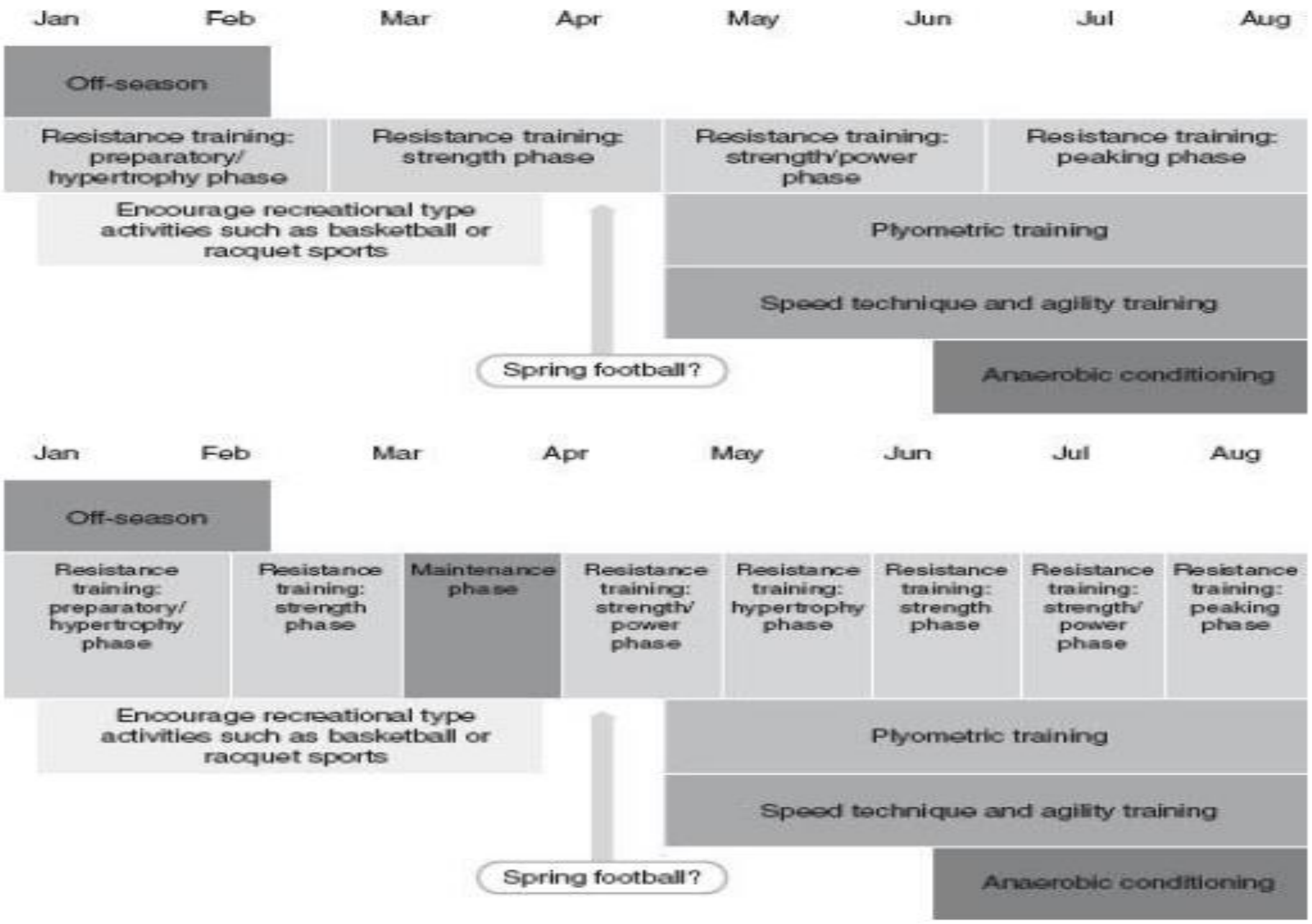

FIGURE 12.1 Two off-season training programs for a strength/power sport, such as American football: (*a*) Does not allow for interruptions in the training program, but (*b*) makes allowances for spring competition.

Hoffman, J. *NSCA´s Guide to Program Design*. Human Kinetics (2012).

MACRO.

Table 2

General resistance training variables during the 25-week training macrocycle.*

Training periods	Weeks	Resistance training (2–3 training sessions per week): Structural exercises — Bench press, squat, dead lifts, and shoulders press — Sets × reps (% of 1-RM)	Assistance exercises — Dumbbells, resistance machines, and functional equipment — Sets × reps (% of 1-RM)	Olympic weightlifting — Snatch, clean and jerk, and various derivatives — Sets × reps (% of 1-RM)
T1 measurements	0	Transition—baseline measurements (T1)		
Hypertrophy and maximum strength	1	4 × 10 – (75% RM)	3 × 12 – (75% RM)	5 × 6 – (75% RM)
	2	4 × 10 – (75% RM)	3 × 12 – (75% RM)	5 × 5 – (80% RM)
	3	4 × 8 – (80% RM)	3 × 10 – (77% RM)	4 × 4 – (85% RM)
	4	4–5 × 8 – (80% RM)	4 × 8 – (80% RM)	4 × 4 – (85% RM)
	5	4 × 8 – (75% RM)	4 × 8 – (80% RM)	5 × 5 – (80% RM)
	6	4–5 × 6 – (85% RM)	4 × 8 – (80% RM)	5 × 4 – (85% RM)
	7	4–5 × 6 – (85% RM)	4 × 10 – (77% RM)	4 × 3 – (90% RM)
	8	4 × 4 – (90% RM)	4 × 8–10 – (80% RM)	5 × 2 – (90% RM)
	9	4 × 4 – (90% RM)	4 × 8–10 – (80% RM)	5 × 2 – (90% RM)
	10	4 × 3 – (85% RM)	4 × 8 – (80% RM)	4 × 2 – (85% RM)
	11	3 × 3 – (75–85% RM) or 4 × 6—(30–40% RM)	2 × 10 – (80% RM)	3 × 2 – (80% RM)
T2 measurements	12	Competitions—second measurements (T2)		
Maximum strength and power	13	4 × 6 – (85% RM)	3 × 8 – (80% RM)	4 × 3 – (90% RM)
	14	4 × 4 – (90% RM)	4 × 8-10 – (80% RM)	4 × 2 – (95% RM)
	15	4 × 4 – (90% RM)	4 × 6-8 – (85% RM)	4 × 2 – (95% RM)
	16	4–5 × 4 – (90% RM)	4 × 6-8 – (85% RM)	5 × 1 – (100% RM)
	17	4 × 6 – (85% RM)	3 × 10 – (80% RM)	4 × 4 – (85% RM)
	18	5 × 4 – (90% RM)	5 × 6-8 – (90% RM)	5 × 2 – (95% RM)
	19	4–5 × 2 – (95% RM)	4 × 6-8 – (95% RM)	5 × 2 – (95% RM)
	20	4–5 × 2 – (95% RM)	4 × 6-8 – (95% RM)	4 × 1 – (100% RM)
	21	3 × 1 – (100% RM)	3 × 8 – (80% RM)	4 × 1 – (100% RM)
	22	3 × 3 – (90% RM)	3 × 8 – (80% RM)	4 × 3 – (90% RM)
Power and speed	23	3 × 3 – (75–85% RM) or 4 × 6 – (30–40% RM)	2 × 10 – (80% RM)	4 × 2 – (80–85% RM)
	24	3 × 3 – (75–85% RM) or 4 × 6 – (30–40% RM)	2 × 10 – (80% RM)	4 × 2 – (80–85% RM)
T3 measurements	25	National outdoor championship—final measurements (T3)		

*RM = repetition maximum, Reps = repetitions.

Table 3

General training program of plyometrics and throwing training during the 25-week macrocycle.*

Training periods	Weeks	Plyometrics (2–3 training sessions per week) — Hurdles, long jumps, box jumps, etc.	Track and field throws (4–5 training sessions per week): 0.5–2 kg heavier	0.5–1.5 kg lighter	Competitive	Shot put, kettlebells, and medicine ball throws (4–5 training sessions per week) — Underhand and backward, upward throws, chest throws, etc.
T1 measurements	0	Transition—baseline measurements (T1)				
Hypertrophy and maximum strength	1	160–200	20–30	—	30–40	60–70
	2	160–200	20–30	—	30–40	60–70
	3	180–220	30–40	—	40–50	70–80
	4	180–220	30–40	—	40–50	70–80
	5	160–200	20–30	—	30–40	60–70
	6	220–240	30–40	—	50–60	80–100
	7	220–240	30–40	—	50–60	80–100
	8	250–270	30–40	—	50–60	100–120
	9	250–270	40–50	—	50–60	120–140
	10	200–220	30–40	—	40–50	100–120
	11	160–200	—	—	30–40	60–70
T2 measurements	12	Competitions—second measurements (T2)				
Maximum strength and power	13	180–220	20–30	—	30–40	80–100
	14	220–240	30–40	—	40–50	80–100
	15	250–270	40–50	—	50–60	100–120
	16	250–270	40–50	—	50–60	100–120
	17	220–240	30–40	—	30–40	80–100
	18	250–270	20–30	20–30	40–50	120–140
	19	250–270	20–30	30–40	50–60	120–140
	20	220–240	—	30–40	50–60	80–100
	21	200–220	—	40–50	50–60	80–100
	22	180–200	—	30–40	50–60	60–80
Power and speed	23	140–160	—	20–30	30–40	60–80
	24	140–160	—	20–30	30–40	60–70
T3 measurements	25	National outdoor championship—final measurements (T3)				

*Plyometrics and throws are presented here as a summary of ground contacts and number of throws during the training weeks, respectively. All plyometrics ground contacts were performed with the intention of maximum velocity during the concentric phase.

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Anousaki, Eleni1; Zaras, Nikolaos2; Stasinaki, Angeliki-Nikoletta1; Panidi, Ioli1; Terzis, Gerasimos1; Karampatsos, Giorgos1 Effects of a 25-Week Periodized Training Macrocycle on Muscle Strength, Power, Muscle Architecture, and Performance in Well-Trained Track and Field Throwers, Journal of Strength and Conditioning Research: October 2021 - Volume 35 - Issue 10 - p 2728-2736 doi: 10.1519/JSC.0000000000004098